



POLITECNICO
MILANO 1863

Dynamics of Electrical Machines and Drives

Direct Torque Control

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R-FOC vs DTC

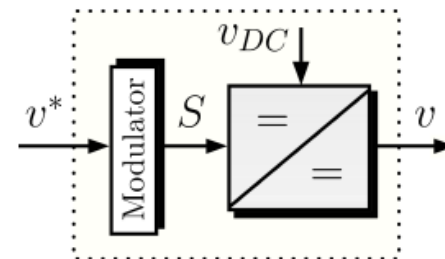
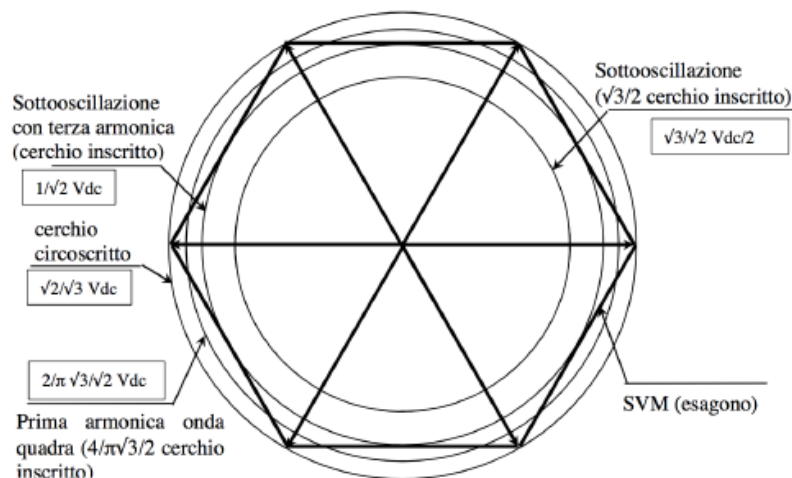
Discussion

- For demanding application, the industry start to ask higher dynamic and more handling to design the controller....
- **Direct Control Approach**
 - ADVANTAGES:
 - higher dynamic (hysteresis controller → **non linear**)
 - ripple managing (smooth → narrow bandwidth)
 - no cross-coupling terms compensation (robust)
 - DISADVANTAGES:
 - variable switching frequency related to the hysteresis band chattering → difficult design of the physical device (variable losses)
 - more powerful hardware (higher sampling time)
 - very sensitive to estimators and motor parameters

Modulator PWM-based

Bottleneck

- Looking inside what we assumed till now as an unitary gain we have:



- The “best” trajectory would be circular one $\rightarrow$ (1st **harmonic only**)
- For the torque it is more important the 1st harmonic, but with the modulation techniques we add also other harmonics (higher order)...
- We can have different modulation techniques (PWM, SVM, square-wave,...)
- From now on we should consider this part!!

Direct Torque Control

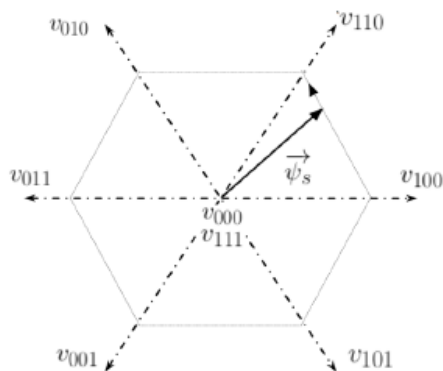
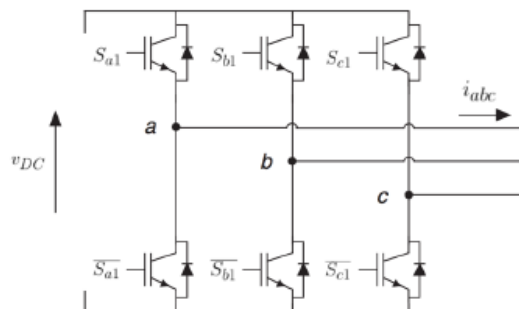
Working Principle (Increment of f_{sw})

- A vector control approach as the R-FOC computes the voltage vector to be applied at the motor as results of the outer loop + the help of a modulator that provide what is requested (e.g. $v^* \rightarrow v$)
 - A direct approach as the DTC doesn't compute the voltage vector as results of a current loop but it selects the *most convenient* voltage vector:
- ① **Number of Voltage Vector?**
 - THREE-PHASE 2L-VSC (we use this one...)
 - 8 vectors (6 active + 2 zero)
 - ② **What does it mean *most convenient*?**
 - The one whose keeps the flux inside a tolerance bandwidth....
 -and it keeps also the torque inside a tolerance bandwidth

Voltage Vectors

Three-Phase 2L-VSC

- The voltage vectors for a three-phase 2L-VSC compose an **hexagon**:



according to the following switches:

voltage vectors v_k	switches		
	S_{a1}	S_{b1}	S_{c1}
v_{000}	0	0	0
v_{100}	1	0	0
v_{110}	1	1	0
v_{010}	0	1	0
v_{011}	0	1	1
v_{001}	0	0	1
v_{101}	1	0	1
v_{111}	1	1	1

- We identify a sector $\theta(n)$ every 60° , for example the sector 1 is defined between -30° and $+30^\circ$ and it contains v_{100}

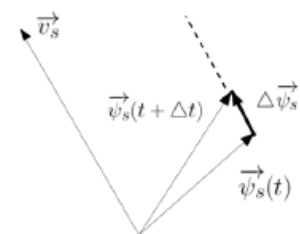
DTC Working Principle #1

Stator

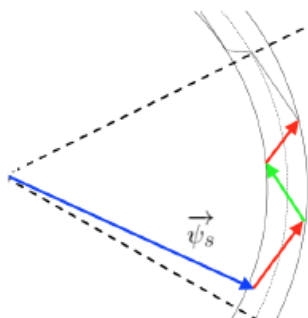
- Each voltage vector produces a movement of the flux that is related to the direction of $\vec{v}_s$

$$\vec{v}_s = R_s \vec{i}_s + \frac{d\vec{\psi}_s}{dt} \quad \Rightarrow \quad \vec{\psi}_s = \int_0^t (\vec{v}_s - R_s \vec{i}_s) dt$$

- In a specific time interval T_s , if the stator resistance is small it follows:



$$\vec{\psi}_s = \vec{v}_s T_s \quad \rightarrow \quad \vec{v}_s = \frac{\Delta \vec{\psi}_s}{\Delta t}$$



- The flux vector's trajectory changes according to the trajectory of the *chosen* voltage vector
- Hence, the flux trajectory is kept in a narrow bandwidth by managing only one voltage vector, regulating his trajectory

DTC Working Principle #3

Flux Control Loop

Increase the Flux || ||

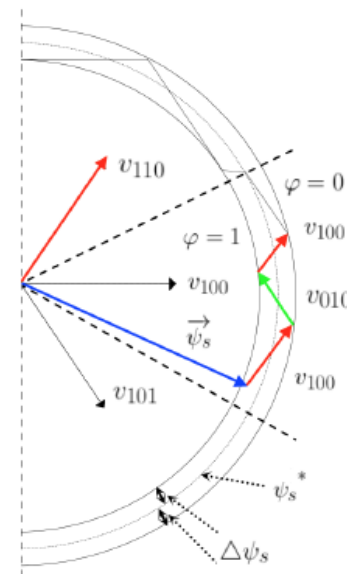
- to increase the amplitude, making a clockwise rotation we use the v -vector of the 1-consequent sector v_{k+1}

Decrease the Flux || ||

- to increase the amplitude, making a counterclockwise rotation we use the v -vector of the 2-consequent sector v_{k+2}

- However, without any other regulation the flux $\vec{\psi}_s$ goes at the maximum speed providing the maximum torque because the angle from the rotor flux is maximum

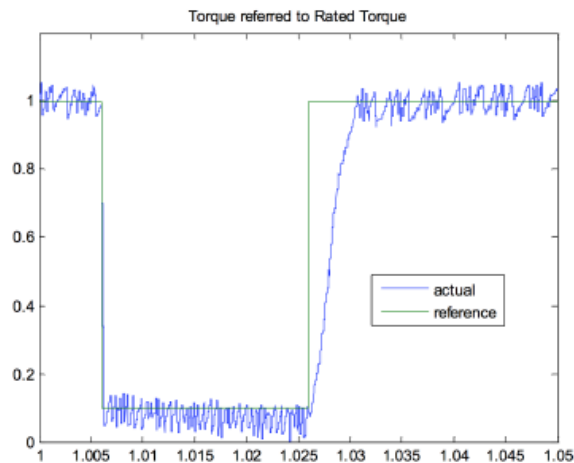
$\Rightarrow$ we need a **Torque Control Loop**



DTC Working Principle #4

Torque Control Loop

- Managing the *sequence* of voltage vector $\vec{v}_s$ we can manage the slip s seen as the angular difference δ between stator and rotor flux
- Hence, changing δ through $\vec{v}_s$ we can arrest or advancing $\vec{\psi}_s$, keeping the torque in a given band.



$$m_e = \frac{n_p}{L_{ks}} \Im(\vec{\psi}_s \vec{\psi}_r) = \frac{n_p}{L_{ks}} |\vec{\psi}_s| |\vec{\psi}_r| \sin \delta$$

- the rotor flux vector moves slowly compared to the stator one (in addition it has a higher inertia)
- The change in the angle δ thus corresponds in good approximation to a change in the angle of the stator flux that is done by managing its advancing trajectory

Look-Up Table and Comparator #1

how to link the flux and torque error

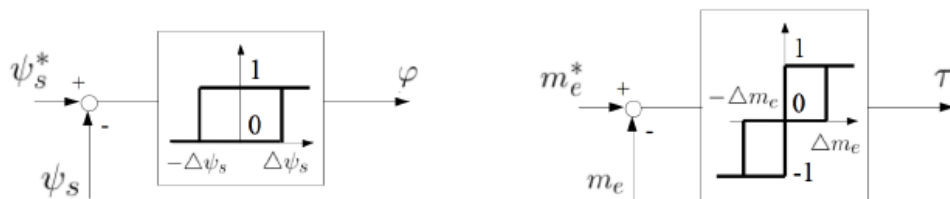
- Stator flux and torque change are both function of $\vec{v}_s$
- As done for the stator flux $\parallel \parallel$ increase/decrease we can link the flux torque relationship as follow:

		torque error τ		
		-1	0	1
flux error φ	1	v_{k-1}	0	v_{k+1}
	0	v_{k-2}	0	v_{k+2}

where

- $\varphi = 1 \rightarrow$ flux increase
- $\varphi = 0 \rightarrow$ flux decrease
- $\tau = 1, -1 \rightarrow$ torque increase (↻, ↺)
- $\tau = 0 \rightarrow$ torque decrease

This is done through **comparators** that evaluate when flux and torque exceed their threshold:



Look-Up Table and Comparator #2

Switching Table

- For Simulink implementation it is easier to build a **switching table** as:

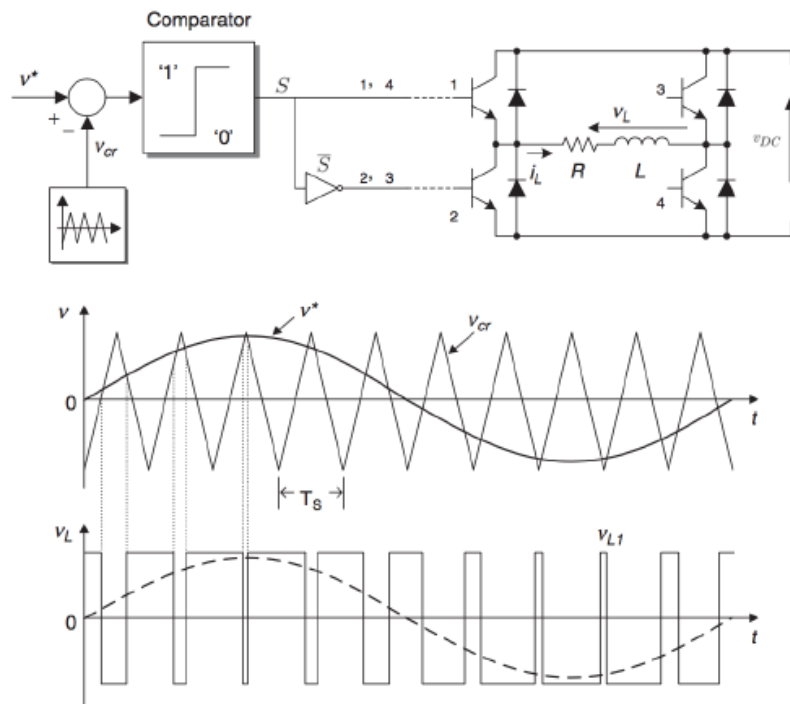
		$\theta(1)$	$\theta(2)$	$\theta(3)$	$\theta(4)$	$\theta(5)$	$\theta(6)$
$\varphi = 0$	$\tau = 1$	v_{010}	v_{011}	v_{001}	v_{101}	v_{100}	v_{110}
	$\tau = 0$	v_{000}	v_{111}	v_{000}	v_{111}	v_{000}	v_{111}
	$\tau = -1$	v_{001}	v_{101}	v_{100}	v_{110}	v_{010}	v_{011}
$\varphi = 1$	$\tau = 1$	v_{110}	v_{010}	v_{011}	v_{001}	v_{101}	v_{100}
	$\tau = 0$	v_{111}	v_{000}	v_{111}	v_{000}	v_{111}	v_{000}
	$\tau = -1$	v_{101}	v_{100}	v_{110}	v_{010}	v_{011}	v_{001}

- This one would be built in two steps:
 - use a Look-Up table 3D $\rightarrow$ **input:** $\tau, \varphi, \theta(n)$ **output:** *integer* of v_k
 - use a selector to link the integer of v_k with a specific switching configuration

Converter Modeling

Single-Phase PWM

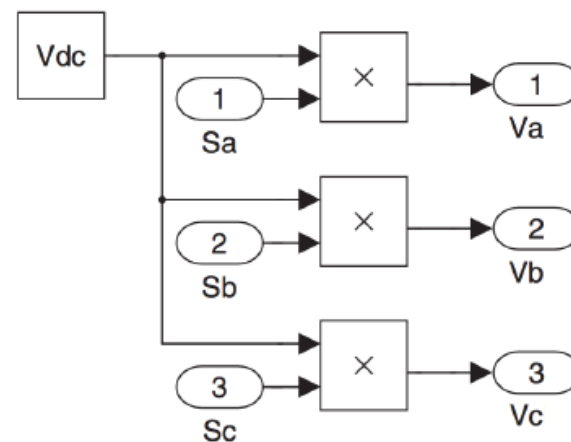
- PWM working principle:



1st harmonic + higher harmonics
($f_{sw} \approx 10kHz$)

The easiest way to implement a converter in simulink is:

$$\underbrace{[v_a \ v_b \ v_c]^T}_{v_{abc}} = \frac{v_{DC}}{2} \underbrace{[S_a \ S_b \ S_c]^T}_{S_{abc} \in (0,1)}$$



Exercise

Electrical Drive for an Air Compressor

Consider the air treatment system of a blast furnace based on a centrifugal compressor (CC) driven by an induction motor with the following characteristics:

- rated voltage 380 V
- rated frequency 50 Hz with pole pairs $n_p = 3$
- stator resistance $R_s = 0.24\Omega$ and inductance $L_s = 59.4\text{ mH}$
- rotor resistance $R_r = 0.175\Omega$ and inductance $L_r = 59.1\text{ mH}$
- mutual inductance $L_m = 57\text{ mH}$

The compressor is coupled to the IM with a gearbox $r = \omega_l / \omega_m = 4$:

- the equivalent inertia seen from the motor side is $J_{eq} = 0.4\text{ kgm}^2$
- friction coefficient $\beta = 0.068$

The compressor behavior see a load torque proportional to the square of the shaft speed as $m_l = k\omega_m^2$ where $k = 0.009\text{ Nm/rad}^2/\text{s}^2$

Exercise

Electrical Drive for an Air Compressor

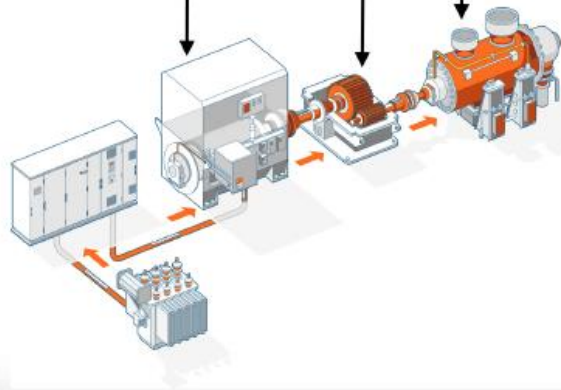
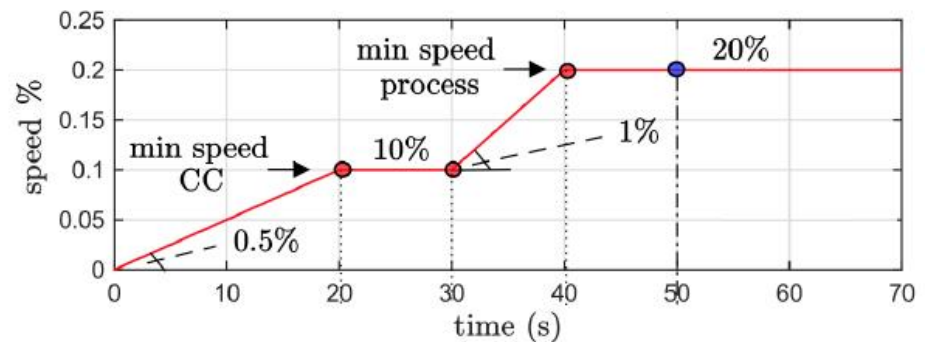


ABB project (Restricted)

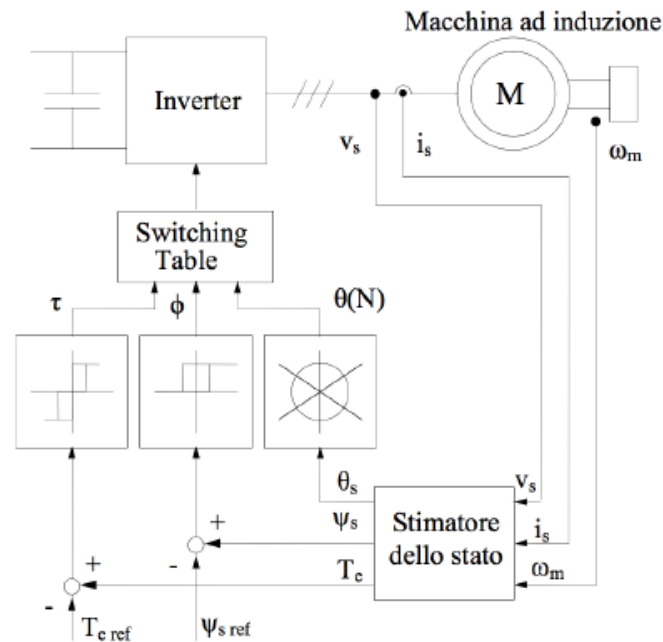
- Design and simulate speed control and a DTC in order to cover the given fluid demand at least one decade faster than the intrinsic characteristic of the uncontrolled system. Figure out:
 - currents, torque, speed profiles compared to the reference one



DTC Scheme of an IM – Simulink Scheme

Functional Scheme

- We have to build this complete scheme:



Remark We must be careful on field coordinates *orientation*